



DEALER BEST PRACTICE SERIES

Managing Fluid Cleanliness

Application Maintenance Site Management Component Rebuild Safety MARC Management

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1.0 Introduction

This Best Practice provides an overview of the importance of maintaining clean fluids and offers suggestions on how best to achieve that. Detailed information can be found in the specific Best Practice publications cited.

Minimizing machine operating cost is critical to minimizing cost-per-ton. Aside from the cost of tires, fuel, and operator, about 70% of total machine operating cost is the life cycle costs of machine powertrain components. On a typical large mining truck, the cost distribution is as follows:

- Engine	40%
- Transmission & Torque Converter	10%
- Final Drive & Differential	40%
- Miscellaneous	10%

Component life cycle cost is roughly defined as cost to rebuild the component divided by actual component life in hours. Example:

(\$100,000 rebuild cost ÷ 10,000 hour life = \$10 hour life cycle cost)

Extending the life of a component is the most important factor in reducing its life cycle cost. This does not mean simply extending overhaul intervals and allowing components to wear more severely. It means implementing a strategy to reduce the rate of wear and achieve longer component life without incurring excessive wear.

2.0 Best Practice Description

The best way to minimize power train cost per hour is to extend component life and utilize the value built into the component. The most effective way to accomplish this is to operate the component with very clean oil throughout its entire life.

Fluid cleanliness management is a strategy to:

- Remove component and/or system break-in debris as quickly as possible on new and rebuilt components.
- Maintain very clean oil in the component and/or system throughout the entire PM interval and entire component life.

3.0 Implementation Steps

3.1 Dealer & Customer Commitment

3.1.1 Understand the Causes of Component Wear and Failure

- Reference [Improving Component Durability](#) booklets (See Section 6.0 *Supporting Attachments*)
- Tolerate Early-Hour Filter Plugging
- Use Off-Board Filtration to Remove Break-In Debris

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3.2 Bulk Fuel Filtration

For a variety of unavoidable reasons, fuel delivered to mine sites is usually contaminated with dirt and water. Because fuel is a very low margin commodity, suppliers almost never provide adequate bulk fuel filtration or exercise recommended storage practices. As a result, mines receive and use contaminated fuel, resulting in premature injector failure and wear out. This results in excessive fuel consumption and often results in mid-life injector set replacement.

The fuel filters on the machine are designed to provide final filtration for moderately clean supply fuel. Machine filtration is not intended to clean fuel contaminated with large amounts of dirt and water. If contaminated fuel is used, the capability of the onboard filtration is overwhelmed and injectors either wear out prematurely or seize.

Bulk fuel filtration has been used in the aviation industry for more than 50 years to address the same problems. Caterpillar has now adopted this proven technology to help mining customers clean contaminated fuel.

Bulk fuel filtration consists of high capacity filters, which remove both excess dirt and water from the supply fuel before it is put into the machine.

Caterpillar has engineered a packaged system to remove both dirt and water. It requires very little maintenance and contains safeguards to prevent contaminated fuel from passing through the unit. The unit is self-contained on a skid and is located between the fuel storage tank and fueling station. It provides single pass filtration, and is offered in four sizes depending on the maximum flow rate of the fuel delivery system.

3.3 Bulk Oil Filtration

A widespread misconception is that new oil is automatically clean because of its appearance. In fact, nearly all new oil is contaminated to some degree with dirt, metal particles, plastic, water, or other foreign debris. These contaminants are introduced in the transportation and storage process from the time the oil leaves the refinery until it is used by the end-user.

Very little new bulk oil meets the recommended Caterpillar cleanliness spec for new oil of ISO16/13. This includes oil delivered in bulk tanks, steel barrels, plastic cubes, and small plastic containers.

Unfiltered new oil should never be taken directly from the container and placed into the machine. This is true whether the oil is being used for refilling a compartment at an oil change interval or simply topping off a system.

A variety of bulk oil filtration methods are available and the best one for each situation is dictated by factors such as: volume of oil used, location, available infrastructure, and cost.

3.4 Off-Board Machine Fluid Filtration

Filtration carts can make a major contribution to extending component life. Many mines and Caterpillar dealers use filtration carts for the major systems during normal preventive

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maintenance. Carts are connected to major systems (rear axle, transmission, hydraulics, steering) and operate unattended while PM services are completed.

3.5 On-Board Machine Fluid Filtration

The micron rating of filters on many mining machine systems are sized so as not to plug during the initial break-in period on new machines. This does not provide optimal filtration capability to maintain very high levels of oil cleanliness after the break-in period is complete.

The most aggressive approach to removing break-in debris as quickly as possible and maintaining the highest level of fluid cleanliness is to use Ultra-High Efficiency (UHE) 6-micron filters in place of standard filters for all machine systems except the engine. Because these filters effectively trap very small particles, some filter plugging will occur during the initial PM periods.

3.6 Breather Filters

Dust entering fluid compartments through inefficient breather filters is often a source of fluid contamination. This can be easily prevented with the use of spin-on High Efficiency 4-micron fuel filters as breathers for all compartments. When used as fuel tank breathers, HE filters have reduced or eliminated premature fuel filter plugging in extremely dusty applications. HE fuel filters are also larger than standard breathers and have much greater capacity.

3.7 Measuring Oil Cleanliness

Component life is maximized when high levels of fluid cleanliness are maintained. The ability to effectively and consistently measure debris in fluids is a basic requirement of managing fluid cleanliness. Tracking fluid particle data is one way to monitor component health. If a particle count raises sharply, an SOS sample can be used to determine the specific wear metal showing elevated levels.

4.0 Benefits

Improved Durability

- Up to 1/3 longer life of powertrain and implement hydraulic components
- Does not apply to engine due to soot in lube oil.

Improved Reliability

- Reduce or eliminate repairs caused by contamination debris.

Improved Parts Reusability

- Reduced wear rates of internal parts improve reusability.

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5.0 Resources Required

- Bulk Fuel Filter Coalescer
- Bulk Oil Filtration
 - o Permanent filtration installation (or)
 - o Portable filtration carts (or)
 - o Barrels (or)
 - o Cubes (or)
 - o On-Machine (or)
 - o Off-Board Filtration Carts
- Portable Particle Counters
- Particle Count Data Management Software
- *Improving Component Durability* Training Booklets

6.0 Supporting Attachments

See [Component Life Management Strategy](#) Document (Click on *Attachments* tab within this document to view attached file) 

Improving Component Durability booklets (see next page)

7.0 Related Best Practices

0808-2.10-1006	-Bulk Fuel Filtration
0808-2.10-1005	-Bulk Oil Filtration
0808-2.10-1002	-Off-board Fluid Filtration
0808-2.10-1002	-On-board Fluid Filtration
0808-2.10-1004	-Breather Filters
0808-2.10-1001	-Measuring Oil Cleanliness

8.0 Acknowledgements

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